



ML Interview Journal #001

POLYNOMIAL REGRESSION

PREPARING FOR DATA SCIENCE INTERVIEWS?

CAN YOU ANSWER THESE
POLYNOMIAL
REGRESSION INTERVIEW
QUESTIONS?



By Naiya Khalid



<https://www.linkedin.com/in/naiya-khalid-510981130/>

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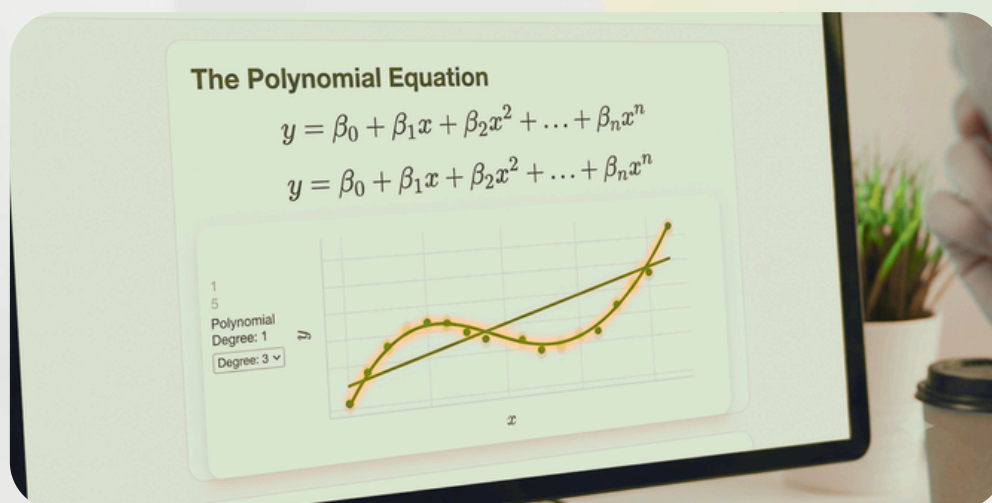
CORE CONCEPTS

WHAT IS POLYNOMIAL REGRESSION?

- EXTENSION OF LINEAR REGRESSION
- MODELS CURVED RELATIONSHIPS
- CREATES POLYNOMIAL FEATURES ($x^1, x^2, \dots$)

WHAT PROBLEM DOES IT SOLVE?

CAPTURES NONLINEAR PATTERNS THAT A STRAIGHT LINE CANNOT MODEL.



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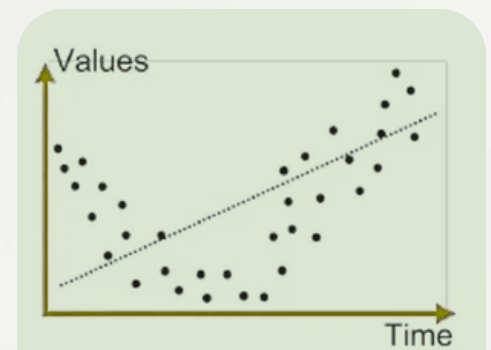




CHOOSING THE RIGHT DEGREE

DEGREE 1

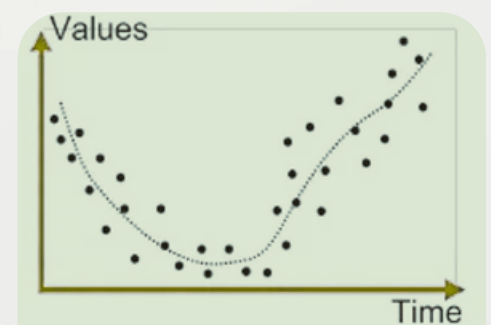
- HIGH BIAS
- UNDERFITTING



Underfitted

DEGREE 2

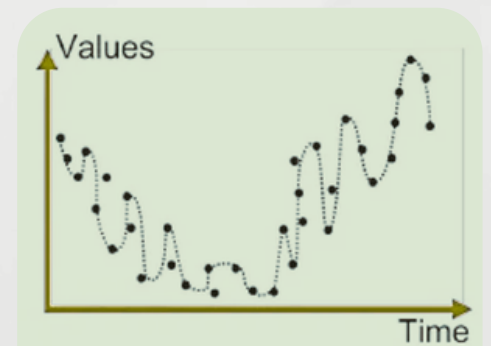
- USUALLY THE BEST BALANCE
- GOOD GENERALIZATION



Good Fit/Robust

VERY HIGH DEGREE

- HIGH VARIANCE
- OVERFITTING



Overfitted

HOW DO WE CHOOSE?



- VALIDATION SET
- CROSS VALIDATION
- RMSE
- R2 SCORE

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BIAS-VARIANCE TRADE-OFF

LOW
DEGREE

- HIGH BIAS
- LOW VARIANCE
- UNDERFITTING



HIGH
DEGREE

- LOW BIAS
- HIGH VARIANCE
- OVERFITTING

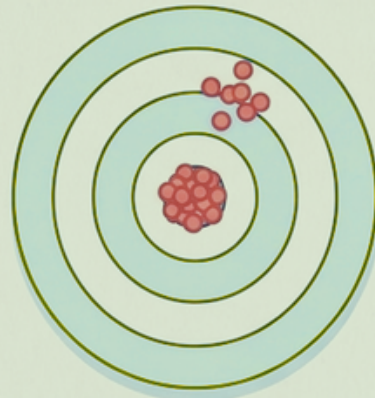


HOW CAN WE REDUCE
OVERFITTING?

- LOWER POLYNOMIAL DEGREE
- MORE TRAINING DATA
- REGULARIZATION



UNDERFITTING

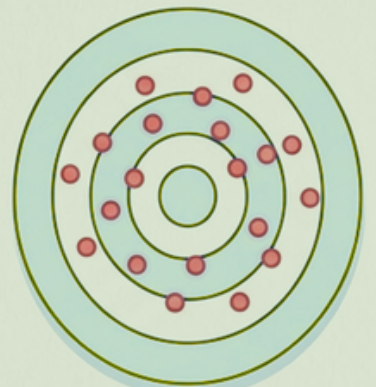


LOW DEGREE
(Model Complexity)



High Bias → Low Variance

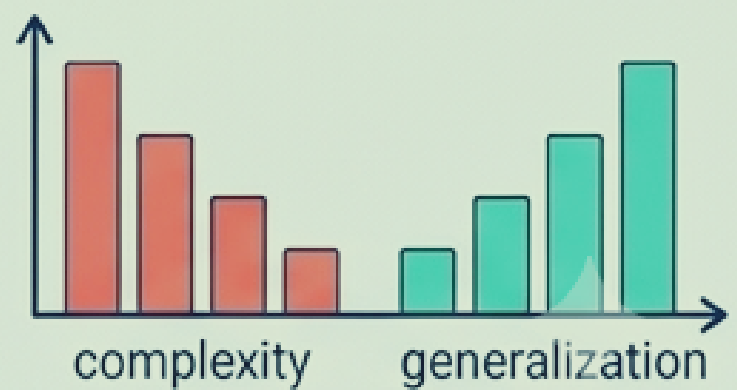
OVERFITTING



HIGH DEGREE
(Model Complexity)



Low Bias → High Variance



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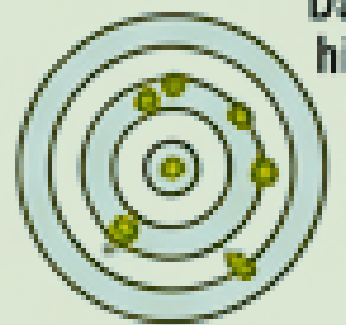




PRACTICAL INTERVIEW QUESTIONS

SHOULD POLYNOMIAL FEATURES BE SCALED?

- ✓ USUALLY YES
 - ESPECIALLY FOR HIGHER-DEGREE MODELS



Variance clusters

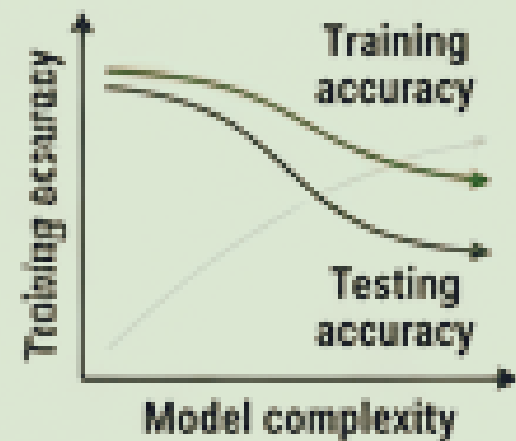
HOW DO YOU DETECT OVERFITTING?



COMPARE:

- ✓ TRAINING PERFORMANCE
- ↓
- ✓ TESTING PERFORMANCE

LARGE GAP = OVERFITTING



WHICH EVALUATION METRICS?



- ✓ RMSE
- ✓ MAE
- ✓ R^2

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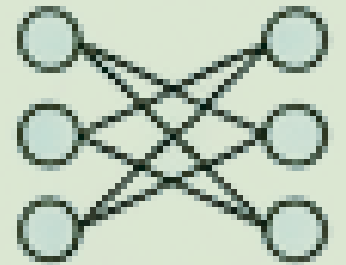


ADVANCED INTERVIEW QUESTIONS

CAN POLYNOMIAL REGRESSION USE MULTIPLE FEATURES?

✓ YES.

IT CAN ALSO CREATE INTERACTION FEATURES.



$x_1 \times x_2$

WHAT ARE INTERACTION TERMS?

EXAMPLE: $x_1 \times x_2$



WHAT IS MULTICOLLINEARITY?

- HIGHLY CORRELATED POLYNOMIAL FEATURES.
- MAY CAUSE UNSTABLE COEFFICIENTS.



WHY USE TRAIN-TEST SPLIT?

- TO EVALUATE
- GENERALIZATION ON UNSEEN DATA.



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REAL-WORLD APPLICATIONS

WHEN SHOULD YOU USE IT?



USE POLYNOMIAL REGRESSION WHEN:

- ✓ RELATIONSHIPS ARE CURVED
- ✓ DATA FOLLOWS SMOOTH NONLINEAR TRENDS

EXAMPLES



SALARY PREDICTION



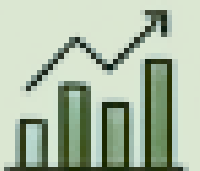
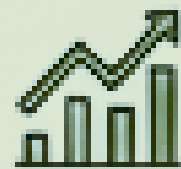
ELECTRICITY DEMAND



MANUFACTURING



POPULATION GROWTH



ADVANTAGES



- ✓ SIMPLE
- ✓ FLEXIBLE
- ✓ INTERPRETABLE

LIMITATIONS



- ✗ OVERFITTING
- ✗ POOR EXTRAPOLATION
- ✗ MULTICOLLINEARITY

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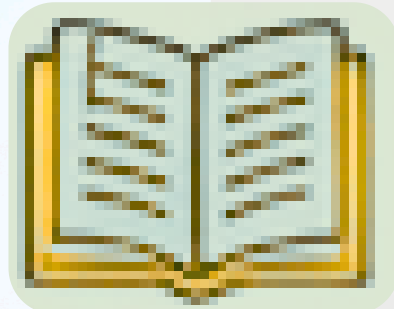
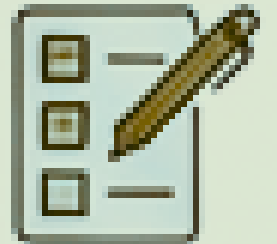


INTERVIEW CHECKLIST

BEFORE YOUR INTERVIEW, ASK YOURSELF:

CAN I EXPLAIN !

- ✓ HOW TO CHOOSE THE DEGREE?
- ✓ HOW TO DETECT OVERFITTING?
- ✓ WHEN TO SCALE FEATURES?
- ✓ WHICH EVALUATION METRICS TO USE?
- ✓ REAL-WORLD APPLICATIONS?
- ✓ ADVANTAGES & LIMITATIONS?



MY TAKEAWAY

I INITIALLY THOUGHT CHOOSING A HIGHER POLYNOMIAL DEGREE WOULD ALWAYS IMPROVE THE MODEL. WHILE STUDYING AND PRACTICING, I LEARNED THAT HIGHER DEGREES CAN EASILY LEAD TO OVERFITTING, SO VALIDATION AND CROSS-VALIDATION ARE ESSENTIAL.

FOLLOW FOR MORE ML INTERVIEW JOURNAL POSTS.

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